

Using and Improving GladLibs

HasMap for Flexible Design

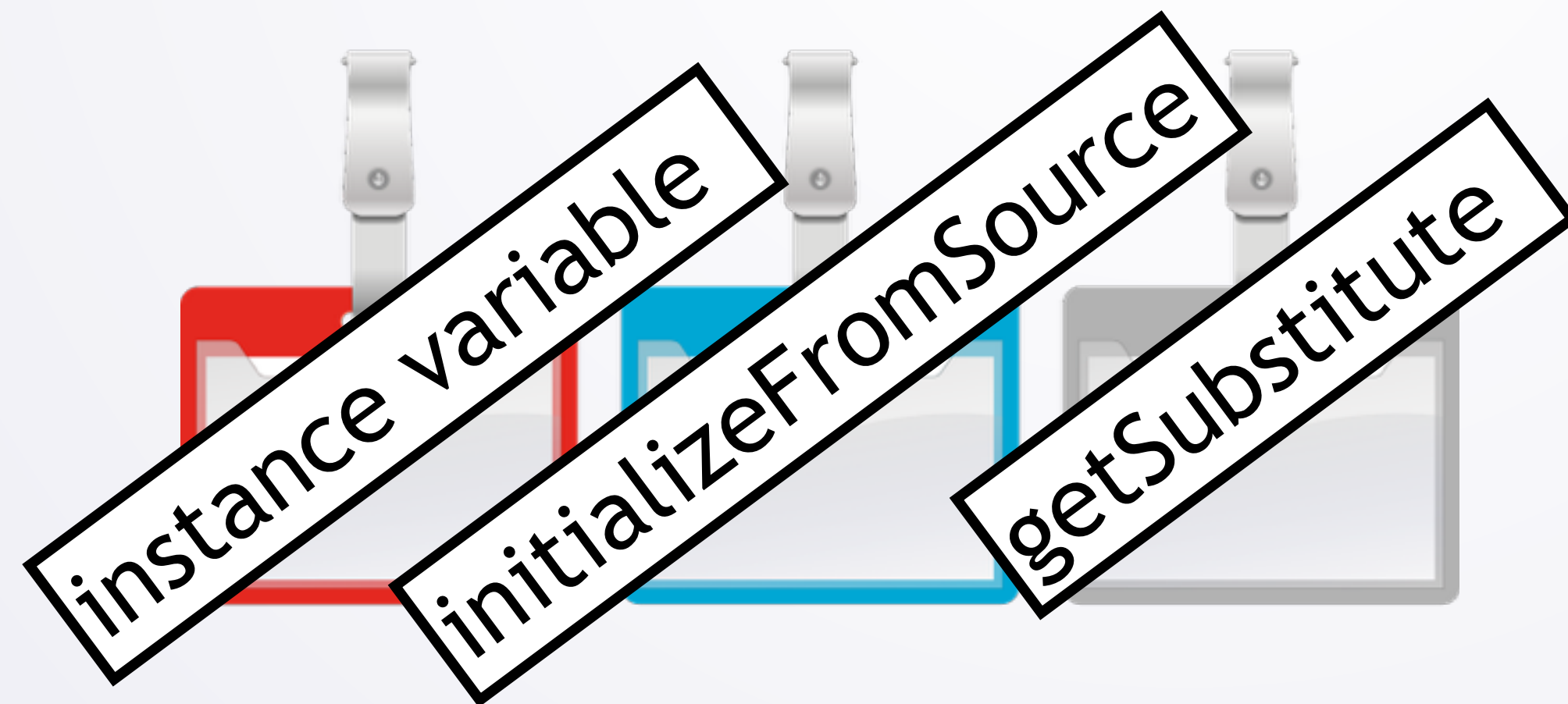
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- To add <verb> label must modify code in several parts of the program
 - Three different parts of the program must be modified
 - Follow conventions: verbList for verb
- Difficult to use different .txt files or URLs, names bound to ArrayLists and labels
 - nounList : noun.txt
 - colorList : color.txt

New Structures for Data and Classes

- Each label is associated with an ArrayList
 - <noun> and nounList, <color> and colorList, ...
- Named instance variables: poor design
 - Define by name, initialize and use by name
- HashMap class helps with flexible design
 - Map or align <label> to ArrayList, no names!
 - Look up the ArrayList given the <label>
 - Like indexOf, but returns ArrayList for <label>

Flexibility in Using HashMap

- Replace seven or more instance variables by one instance variable

```
private ArrayList<String> adjectiveList;  
private ArrayList<String> nounList;  
private ArrayList<String> colorList;  
private ArrayList<String> countryList;  
private ArrayList<String> nameList;  
private ArrayList<String> animalList;  
private ArrayList<String> timeList;
```

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private HashMap<String, ArrayList<String>> myMap;
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Looking up Values in HashMap

- Sequence of if-statements in getSubstitute replaced by one statement with HashMap!

```
private String getSubstitute(String label) {  
    if (label.equals("country")) {  
        return randomFrom(countryList);  
    }  
    if (label.equals("color")){  
        return randomFrom(colorList);  
    }  
    // more code here for labels ...  
  
    if (label.equals("number")){  
        return ""+myRandom.nextInt(50)+5;  
    }  
}
```


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Improvement, but Room for More!

- In GladLib.java must assign values to each named instance variable

```
private void initializeFromSource(String source) {  
    adjectiveList= readIt(source+"/adjective.txt");  
    nounList      = readIt(source+"/noun.txt");  
    colorList     = readIt(source+"/color.txt");  
    countryList   = readIt(source+"/country.txt");  
    nameList      = readIt(source+"/name.txt");  
    animalList    = readIt(source+"/animal.txt");  
    timeList      = readIt(source+"/timeframe.txt");  
}
```


Improvement, but Room for More!

- In GladLib.java must assign values to each named instance variable
 - HashMap is better, filename and label linked

```
private void initializeFromSource(String source) {  
    String[] labels = {"country", "noun", "animal",  
                      "adjective", "name", "color",  
                      "timeframe"};  
    for(String s : labels){  
        ArrayList<String> list = readIt(source+"/"+s+".txt");  
        myMap.put(s, list);  
    }  
}
```


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Finding Nouns and Verbs Anywhere

- File associates labels and replacement info
 - .properties or property file

```
verb:http://gladlibs.com/verbs.txt  
noun:http://gladlibs.com/nouns-funny.txt  
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private void initializeFromSource() {  
    for(String s : myLabelSource.keySet()){  
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